Eyeriss-v2 CNN Accelerator — RTL Architecture Diagrams

Team SMC26-30 | Target: LEGNet on FPGA | Dataflow: Hierarchical Mesh NoC + CSC Encoding

1. System-Level Overview

CNN ACCELERATOR — SYSTEM OVERVIEW

Off-Chip DRAM

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CLUSTER GROUP CONTROLLER

▼	▼	▼
GLB CLUSTER (SRAM Banks)	**ROUTER CLUSTER (iact / weight / psum)**	**PE CLUSTER (9 × 4 = 36 PEs)**
▼	◀ ▶	▲
iact SRAM weight SRAM psum SRAM	iact Router weight Router psum Router	PE Core × 36 (MAC + SPads)

2. Processing Element (PE) Core

PE CORE — INTERNAL ARCHITECTURE

iact Data (8-bit INT)	weight Data (8-bit INT)	psum Data (20-bit)
▼	▼	▼
iact SPad (16 deep)	**weight SPad (9 deep)**	**psum SPad (16 deep)**
▼	▼	▼

MAC DATAPATH (iact × weight + psum → psum)

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FSM: IDLE → LOAD_WEIGHTS → LOAD_IACT → MAC → PSUM_STREAM

▼

psum Output (20-bit signed)

3. Router Cluster

ROUTER CLUSTER

iact ROUTER (per PE row) Modes: UNICAST MULTICAST (×6) HOR_CAST VER_CAST BROADCAST	weight ROUTER (per PE row) Circuit-switching MUX GLB → PE Horiz pass-through Valid/Ready handshake	psum ROUTER (per PE column) PE → accum → GLB North / South links Signed 20-bit Valid/Ready handshake
▼	▼	▼
36 PE Cores (iact distributed)	**36 PE Cores (weights loaded)**	**GLB psum SRAM (results collected)**

4. Global Buffer (GLB) Cluster

GLB CLUSTER (Global Buffer)	
iact SRAM 256-deep × 8-bit 3 read ports (1 per iact router) 1 write port 3-cycle read latency	weight SRAM 256-deep × 8-bit 3 read ports (1 per weight router) 1 write port 3-cycle read latency
▼	▼
psum SRAM 32-deep × 20-bit (signed) 1 write / 1 read port Stores intermediate & final OFM	*Read Protocol: 1. Set read_addr 2. Assert read_en 3. Data valid after 2 cycles 4. read_done pulses at end*

5. Convolution Dataflow

DATAFLOW — IACT-STATIONARY / WEIGHT-STATIONARY HYBRID			
Step	**Action**	**Data Path**	**Data Type**
1	Load weights from GLB	GLB weight SRAM → weight router → weight SPad	8-bit INT (stationary)
2	Stream iacts with multicast reuse	GLB iact SRAM → iact router → iact SPad (shared across rows)	8-bit INT (multicast)
3	MAC operations in each PE	iact SPad × weight SPad → accumulated into psum SPad	20-bit signed accumulator
4	Stream psums to GLB	psum SPad → psum router (accumulated per column) → GLB psum SRAM	20-bit signed output

All signals use valid/ready handshake. Precision: 8-bit INT (iact & weight), 20-bit signed (psum). PE array: 9 rows × 4 cols = 36 PEs per cluster.